

Volatility Surface & Rough Heston Pipeline

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title: "Volatility Surface & Rough Heston Pipeline" subtitle: "Progress report — Sessions D to F" author: "Akin Akandere" date: "13 September 2020"

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Status

All six planned sessions are complete. Since the last report (Sessions A–C, 10 September), your analytics review has been worked through item by item. The Markovian lift of rough Heston is built and validated (Session D); rough Heston is calibrated and compared with vanilla Heston (Session E); and Hawkes jump arrivals and Kalman filtering are each attached to three volatility hosts (Session F).

	Sessions A–C (10 Sep)	now
Sessions complete	3 of 6	6 of 6
Test suites	6	10
Tests passing	274 / 274	487 / 487
Analytical health checks	50 / 50	90 / 90 (29 s)
Production lines	2,268	6,475
Test lines	1,641	3,634
Commits	15	26
Python / numpy	3.12.3 / 2.5.2	3.12.3 / 2.5.2

The repository is local-only. No remote, nothing pushed.

Three things in this report overturn or qualify earlier statements, and they are collected in *Corrections* near the end rather than scattered: a stability claim made in Session D, a headline number from Phase 1, and several plan targets that turned out to be unattainable at the stated sample sizes.

The analytics review

Each item from your review, with what was done. In one case your proposed method was implemented and measured to fail, and something else was shipped.

Heston characteristic function as $\xi \rightarrow 0$. Done, as exact identities rather than only a Taylor branch. $(a-d)(a+d) = -\xi^2(u^2+iu)$ makes $(a-d)/\xi^2$ exact with no cancellation. The logarithm becomes $\log 1p$ of an $O(\xi^2)$ quantity, evaluated by an accurate complex $\log 1p$ (numpy's is not accurate for small arguments), with your Taylor series below $\xi = 1e-4$. The two branches agree to $3e-16$. Against the exact Black–Scholes characteristic function the error is now **2e-16 at $\xi = 1e-8$, 1e-10 and 1e-12**, where the literal form is 19%, 100% and 100% wrong. The literal form is kept as `char_func_naive` so the cancellation is measured, not described.

Second derivatives on non-uniform strike grids. Your proposal — map the grid through a coordinate $x = f(K)$, differentiate uniformly in x , chain-rule back — was implemented and measured (`debug_fd_methods.py`, orders from doubling against the exact lognormal density):

grid	3-point	mapped 3-point	log-map	Fornberg 5-point
smooth stretch	2.00	2.00	1.99	3.99
geometric	2.00	2.00	2.00	3.98
\$1 / \$5 kink (SPY-like)	1.07	0.01	0.87	3.06
random spacing	0.84	-0.12	0.84	2.76

The mapping is second order only when $K(x)$ is itself smooth. At a kink its numerically differenced metric K'' is $O(1)$ wrong, and the method does not converge at all. On smooth grids the plain stencil is already $O(h^2)$, because $h_2 - h_1$ is $O(h^2)$ there. Shipped instead: a five-point local polynomial (Fornberg weights), exact for quartics on any grid. The Breeden–Litzenberger error against the analytic density fell from $7.9e-8$ to $5.3e-12$. The $2.08e-11$ the old health check reported was rounding (floor $5.1e-11$), not truncation, and the check now says so.

Density non-negativity. `density_for_sampling` floors at machine epsilon, renormalises to unit mass, and reports how much negative mass it removed. It is a separate function on purpose: the raw density keeps its sign, because a negative density is the arbitrage signal the panel exists to show.

κ identifiability. The 50% threshold was not tightened. The calibrator takes a Tikhonov prior, in units of “a 100% deviation costs as much as λ quotes each one standard error off”. The prior's cost is reported separately from the data cost, so a prior that disagrees with the market is visible. Under 0.5 vol-point noise the κ spread across seeds fell from 34.7% to 0.07% with unchanged fit quality. `kappa_from_variance_swap` supplies the external pin you suggested.

Monitoring over runs. Every health-check run is appended to a history, and `--trend` reports mean, standard deviation and a drift z-score per check, with your rules for the Jacobian condition number (median < 100 , spikes $> 10^4$) and the clean-data RMSE (rolling mean $< 5e-9$). The RMSE rule was first written $100\times$ too loose and has been fixed. On its first real run the drift monitor flagged the rough-objective speed-up ($0.94\text{ s} \rightarrow 0.60\text{ s}$, $z = -21$), a genuine step change, correctly caught.

The fBm fixture bug (`docs/fbm_helper_bug.md`, reproduced by `debug_fbm_helper.py`, debugged once for your pass).

- **Symptom.** The Session A helper sampled the singular kernel $(t-s)^{H-\frac{1}{2}}$ at the right end of each cell, capping the weight of the newest shock at 1. Increments came out with lag-1 autocorrelation -0.18 against the -0.41 fractional Gaussian noise requires, and the structure function read **$H = 0.26$ for a planted 0.12** .
- **Diagnosis.** Using cell means instead gets to 0.17. Treating the singular first cell exactly (the Bennedsen–Lunde–Pakkanen hybrid scheme) gets 0.1207. It is not the non-stationary start: dropping 20,000 samples moves the estimate by 0.0004.

Scope correction. Hawkes arrivals, the Kalman filter and the dual Kalman filter are not OU-specific. Each is built once and run on the OU, classical Heston and lifted rough Heston hosts. Corrected in the previous report and in the plan.

Numerical methods

New in Sessions D–F

Completely monotone kernel and its lift. $K(t) = t^{\alpha-1}/\Gamma(\alpha)$, $\alpha = H + 1/2$, is the Laplace transform of $\mu(dx) = x^{\alpha-1}/(\Gamma(\alpha)\Gamma(1-\alpha)) dx$. A quadrature of μ by cell mass and cell mean (the Abi Jaber–El Euch construction) gives $K(t) \approx \sum w_i e^{(-x_i t)}$. The first cell starts at zero, so the long-memory mass is kept. The range $[0.1, 10^5]$ with $N = 24$ reproduces K to **0.87% on [1 hour, 2 years]**; the published $r_n = 1 + 10n^{-(0.9)}$ rule measured 27% on the same range. Each factor $U_i = \int e^{(-x_i(t-s))} dz$ is an OU process, so the rough variance becomes a 24-dimensional Markov system.

Lifted Riccati system. The characteristic function is exp of the solution of N coupled Riccati ODEs, derived by Itô on the affine ansatz. With $N = 1$ and $x = 0$ it is Heston's own Riccati pair. Two independent time-steppers solve it:

- **ETDRK4** (Cox–Matthews), fourth order: measured 3.99–4.05 against closed-form Heston.
- **Exponential trapezoidal rule, implicit in the non-linearity.** The implicit equation is a scalar quadratic per u with a closed-form root. Second order, fourth with Richardson extrapolation.

Both treat the linear part exactly, so the fast factors (x up to 10^5) cost nothing. The φ -functions use their series below $|c| = 0.1$, where the direct formulas cancel.

Stability constants, measured. Both schemes are stable only below a step set by $z = h^{\alpha}/\Gamma(1+\alpha) \cdot (\xi(1+|\rho|)|u| + \kappa)$: ETDRK4 at about 2.65, the implicit scheme between 15 and 25. Step counts are sized from these (2 and 12 used). The rough kernel makes both far stricter than for Heston, because $h^{\alpha} \gg h$ at small h .

Lewis contour with checked solves. The rough pricer evaluates the characteristic function on $\text{Im } u = -1/2$, which needs only $E[S^{1/2}]$, finite for every martingale model. That makes $|\phi(u - i/2)| \leq 1$ a hard invariant, and every solve is checked against it. The difference between the M - and $2M$ -step solves serves as an instability detector: a stable solve differs at $\sim 1e-7$, an unstable one at $O(1)$. The u -grid uses Gauss–Legendre panels that double in width ($2,176 \rightarrow 640$ nodes at one day), and the tail is read at the cut-off.

Implementation detail that mattered. OpenBLAS spent about 1.3 ms per ODE step starting threads for $24 \times n$ matrix products. The contractions now use einsum below 1,000 nodes and matmul above, and the rank-3 update is a single pre-stacked real-times-complex product. Result: 6–13× faster per step.

Objective with honest failure. Quotes the model cannot price are charged at the implied-vol ceiling (500%), so failing can never lower the cost. Quotes below resolution in *both* market and model still contribute zero.

Standard errors of calibrated parameters. Gauss–Newton covariance at the truth, in sandwich form when the fit weights and the noise disagree. This is what answered “is H identified by this surface?”

Hawkes process.

- **Simulation:** exact Ogata thinning, vectorised across paths, with planted events and burn-in to stationarity.
- **Closed forms:** stationary intensity $\mu/(1-\alpha/\beta)$; count variance from Hawkes (1971); expected intensity after planted events, whose excess decays at $\beta - \alpha$.
- **Estimation:** maximum likelihood by the $O(n)$ recursion (Ozaki), with observed-information standard errors.
- **Goodness of fit:** the time-rescaling theorem with a Kolmogorov–Smirnov test.

Exact expectations for jump hosts. Every host's conditional mean is linear in its state and in the arrivals, and expected events per step have a closed form. So the noise-free discrete update gives the Monte Carlo's expectation exactly: no discretisation gap between simulation and analysis.

Kalman filtering on three state spaces. One linear-in-the-state filter (Joseph form):

- **OU:** your lecture's filter exactly.
- **CIR:** state-dependent process noise, i.e. quasi-maximum-likelihood.

- **Lifted rough Heston:** the 24 factors as the state, with rank-one process noise $\begin{bmatrix} \xi^2 & \mathbf{V} & \Delta t & \mathbf{g} & \mathbf{g}^\top \end{bmatrix}$.

A pure-float path for one-dimensional states is 25× faster and identical to rounding.

Adaptive filtering. Additive covariance matching along the direction shocks enter the state. The multiplicative version divided by an innovation variance that collapses to 10^{-20} when the rough variance touches zero. A persistence-gated robust variant treats one large innovation as a bad print (Huber-style) and a run of same-signed ones as a regime change.

Parameter learning, four routes.

- **Offline:** maximum likelihood through the filter.
- **Online, Wan–Nelson:** the dual EKF with recurrent derivatives.
- **Online, joint:** the augmented-state EKF.
- **Online, recursive MLE:** full sensitivity equations for $\mathbf{x}$, $\mathbf{P}$ and the gain, with Gauss–Newton steps on the innovation likelihood. This is the correction Ljung (1979) showed is needed for consistency.

Forecast comparison. Diebold–Mariano with Newey–West variance.

Carried over from Sessions A–C

Black-76 in the forward measure; implied vol by Newton on vega with a bisection guard; forward from put–call parity regression; raw SVI by the Zeliade reduction; the Heston characteristic function with the Albrecher branch; Lewis and Carr–Madan with composite Gauss–Legendre; Levenberg–Marquardt with bounded-logistic transforms; Breeden–Litzenberger; the structure function and $\zeta(q) = qH$; full-truncation Euler; Davies–Harte circulant embedding. Details in the previous report.

Planned or open

- Learning H itself inside the filter. H defines the lifted factors, so a parameter filter that moves it changes what the state means.
- A positivity-preserving scheme for rough variance. The lifted Euler scheme lets V go below zero, and that biases QML κ (see *Key results*).
- Hawkes estimation from real jump times, and the nearly unstable Hawkes $\rightarrow$ rough volatility limit as a model rather than a citation.
- eSSVI, arbitrage-free by construction across slices.

Literature and resources

Methods below were implemented from their published constructions. Apart from Shevchenko (2014), retrieved on 10 September, the papers were not downloaded into `docs/reference`.

Used in Sessions D–F

- Abi Jaber & El Euch, *Multifactor approximation of rough volatility models*, SIAM J. Financial Math. (2019): cell-mass / cell-mean nodes for the lift.
- El Euch & Rosenbaum, *The characteristic function of rough Heston models*, Math. Finance (2019): the fractional Riccati equation the lifted system approximates.
- Cox & Matthews, *Exponential time differencing for stiff systems*, J. Comput. Phys. (2002): ETDK4.
- Lewis, *A simple option formula for general jump-diffusion and other exponential Lévy processes* (2001): the $\text{Im } u = -\frac{1}{2}$ contour.
- Andersen & Piterbarg, *Moment explosions in stochastic volatility models*, Finance & Stochastics (2007): why Carr–Madan’s damped contour failed during a fit.
- Fornberg, *Generation of finite difference formulas on arbitrarily spaced grids*, Math. Comp. (1988).
- Bennedsen, Lunde & Pakkanen, *Hybrid scheme for Brownian semistationary processes*, Finance & Stochastics (2017): the exact first cell in the fBm bug diagnosis.
- Hawkes, *Spectra of some self-exciting and mutually exciting point processes*, Biometrika (1971): the count variance.
- Ogata, *On Lewis’ simulation method for point processes*, IEEE Trans. Inf. Theory (1981): thinning.
- Ozaki, *Maximum likelihood estimation of Hawkes’ self-exciting point processes*, Ann. Inst. Stat. Math. (1979).
- Brown, Barbieri, Ventura, Kass & Frank, *The time-rescaling theorem and its application to neural spike train data analysis*, Neural Computation (2002).
- Kalman (1960); your lecture notes for the OU state space, the gain, and strict versus adaptive filtering.
- Mehra, *Approaches to adaptive filtering*, IEEE TAC (1972): covariance matching.
- Wan & Nelson, *Dual extended Kalman filter methods*, in Haykin (ed.), *Kalman Filtering and Neural Networks* (2001).
- Ljung, *Asymptotic behavior of the extended Kalman filter as a parameter estimator for linear systems*, IEEE TAC (1979), and Ljung & Söderström, *Theory and Practice of Recursive Identification* (1983).
- Diebold & Mariano, *Comparing predictive accuracy*, JBES (1995); Newey & West (1987).
- Shevchenko, *Fractional Brownian motion in a nutshell* (2014): Wood–Chan positive definiteness, and the filtered-variation H estimator (not yet implemented).

To be used

- El Euch, Fukasawa & Rosenbaum, *The microstructural foundations of leverage effect and rough volatility*, Finance & Stochastics (2018), and Jaisson & Rosenbaum (2016). Nearly unstable Hawkes processes converge to rough volatility, which is the theoretical counterpart of the Session F finding that the second-spike property on rough Heston needs near-critical clustering.
- Aït-Sahalia, Cacho-Diaz & Laeven, *Modeling financial contagion using mutually exciting jump processes*, JFE (2015): Hawkes jumps in asset prices, for estimation on real data.
- de Jong (2000) and Chen & Scott (2003) on quasi-maximum-likelihood Kalman filtering of square-root models, relevant to the positivity-floor bias.
- Bayer, Friz & Gatheral, *Pricing under rough volatility* (2016).

Health and sanity checks

Run with `python healthcheck.py` (`--trend` adds the run history). The suites assert that behaviour is correct; this reports the numbers behind those assertions.

Foundations

check	measured	threshold	verdict
norm_pdf vs scipy	0	1.00e-15	PASS
norm_cdf vs scipy	2.22e-16	1.00e-15	PASS
put-call parity C-P = F-K	2.84e-14	1.00e-09	PASS
vega vs central difference	4.23e-09	1.00e-04	PASS
implied-vol round trip	8.05e-16	1.00e-06	PASS
fd_first exact on a quadratic	7.46e-14	1.00e-12	PASS
fd_second on a quadratic vs its rounding floor 2.08e-11 measured, floor 5.06e-11	0.41 x floor	1 x floor	PASS
shipped 2nd derivative: order on a \$1/\$5 kink grid Fornberg 5-point; mapped-coordinate idea measured at 0.01	3.077	2	PASS
3-point stencil order on the same grid (kept for contrast) the documented first-order loss	1.086	1.5	PASS

Heston cf

check	measured	threshold	verdict
phi(0) = 1	0	1.00e-12	PASS
phi(-i) = 1 (martingale)	0	1.00e-10	PASS
phi(-u) = conj(phi(u))	0	1.00e-12	PASS
phi(u) <= 1	0.9999	1	PASS
branch continuity, shipped (max/median step) Albrecher g=(a-d)/(a+d)	5.242	50	PASS
branch discontinuity, reciprocal form the trap, kept to prove it is real	2825	500	PASS
trap / shipped separation	538.9 x	100 x	PASS
xi -> 0 degenerates to Black-Scholes at xi=1e-8, cancellation-free form; was 5.9e-09 at xi=1e-4	2.22e-16	1.00e-14	PASS
naive-form cancellation at xi=1e-6 (kept to prove it) shipped form is 5.7e-13 at the same xi	3.44e-05	1.00e-06	PASS

Pricers

Volatility Surface & Rough Heston Pipeline

check	measured	threshold	verdict
Lewis vs exact Black-76	3.55e-15	1.00e-10	PASS
Carr-Madan vs exact Black-76	1.94e-16	1.00e-10	PASS
Lewis vs Carr-Madan on Heston independent routes, 5 maturities	4.99e-15	1.00e-08	PASS
Carr-Madan insensitive to damping alpha alpha 0.75 to 3.0	6.11e-16	1.00e-08	PASS
Fourier vs Monte Carlo (sigmas) residual is Euler O(dt) bias	0.3618 sd	5 sd	PASS
simulated $E[S/F] = 1$	1.94e-06	6.12e-04	PASS
risk-neutral density mass	1	1	PASS
risk-neutral density mean = forward	1	1	PASS
density non-negativity (min/peak) tail dither scales as $1/dK^2$	-8.87e-10	-1.00e-07	PASS
sampling density: min $\geq \epsilon$ and mass = 1 floored 622 pts, removed -5.1e-10 of mass	0	1.00e-12	PASS

Estimators

check	measured	threshold	verdict
H from ATM skew (planted 0.12) $r^2 = 1.00000$	0.12	0.12	PASS
H from structure function, worst of 4 planted $H = 0.10, 0.12, 0.30, 0.50$	0.0062	0.03	PASS
two independent H routes agree surface route vs path route	0.002891	0.05	PASS
forward from put-call parity $r^2 = 1.000000$, no rate or dividend input	1.14e-13	1.00e-06	PASS
discount factor from parity	4.44e-16	1.00e-09	PASS
SVI parameter recovery	1.43e-08	0.001	PASS
SVI slice passes Durrleman $g_{\min}$ over data = 0.2297	1	1	PASS
butterfly audit: false positives on a convex slice	0	0	PASS
butterfly audit: catches a planted dent flags the dent's neighbours	2	1	PASS
calendar audit: clean on monotone w	0	0	PASS
calendar audit: catches a reversal	3	1	PASS

Calibration

Volatility Surface & Rough Heston Pipeline

check	measured	threshold	verdict
plant-and-recover, worst relative error 7 iters, 0.3s	3.13e-11 rel	0.01 rel	PASS
fitted RMSE on clean data	4.06e-09 vol pts	0.001 vol pts	PASS
Jacobian condition number at the solution high = parameters trade off; see identifiability	30.05	1.00e+08	PASS
spread of v0 under 0.5vp noise stable	3.775 %	10 %	PASS
spread of kappa under 0.5vp noise unidentified	34.73 %	50 %	PASS
spread of theta under 0.5vp noise stable	2.803 %	50 %	PASS
spread of xi under 0.5vp noise loose	19.13 %	50 %	PASS
spread of rho under 0.5vp noise stable	3.73 %	10 %	PASS
spread of kappa with a Tikhonov prior (weight 50) prior from a history or a variance swap	0.07281 %	10 %	PASS

Markovian lift

check	measured	threshold	verdict
$K(t) = \int e^{-\lambda t} \mu(dx)$ representation $\mu(dx) = x^a / (\Gamma(a)\Gamma(1-a)) dx$, $a = H+1/2$	6.67e-10 rel	1.00e-08 rel	PASS
kernel error, N=24, t in [1 day, 2 y] max relative; H = 0.12	0.008697 rel	0.01 rel	PASS
kernel error on [1 hour, 2 y] a one-day option lives here; 12% below 5 minutes	0.008697 rel	0.01 rel	PASS
Laplace-domain error, z in [0.1, 100] (plan: 1%, see note) 3.5% is the memory beyond 2 y; 1% is not reachable with a 2 y fit	0.03508 rel	0.05 rel	PASS
Laplace-domain error on the pricing band z in [1, 100] a year down to a few days	0.01419 rel	0.03 rel	PASS
lifted recursion == SOE convolution (identity)	4.05e-14	1.00e-12	PASS
lifted path vs true-kernel Volterra path (L2, same noise)	0.008001 rel	0.01 rel	PASS
N=1 at x=0 reproduces closed-form Heston (ETDRK4, 200 steps)	7.09e-10	1.00e-09	PASS
full N-node lift at H = 0.4999 vs Heston shrinks 10x per decade of (1/2 - H)	1.64e-05	1.00e-04	PASS
ETDRK4 vs implicit trapezoidal + Richardson (H=0.12) independent time-steppers, 1 d / 30 d / 1 y	3.79e-06	5.00e-06	PASS
rough objective evaluation, 10 expiries x 13 strikes auto scheme; vanilla Heston is ~14 ms	0.5893 s	3 s	PASS

Rough robustness

check	measured	threshold	verdict
extrap at 120 steps, u to 1200 (NOT unconditionally stable) phi(u - i/2) must be <= 1; kept to prove the constraint	1.21e+36	1	PASS
stability-sized extrap, same u range: max phi(u - i/2) 383 steps	0.9988	1	PASS
runaway parameters: maturities priced in-band (of 4) Carr-Madan returned 1.9e18 / 2.9e28 at 30 / 90 d here	4	4	PASS
unpriceable quote: minimum charge (vol) vs 0 before exact fit costs 1.4e-22	4.754	4.754	PASS

Rough identifiability

check	measured	threshold	verdict
SE(H), 20 quotes, equal weights, 0.5 vp noise	0.04711	0.06	PASS
SE(H), same quotes, vega^2 weights H not identified: the weights discard the short end	0.2024	0.15	PASS
corr(H, xi) at the truth the rough analogue of kappa/theta	0.9776	0.9	PASS

Hawkes

check	measured	threshold	verdict
mean rate vs mu/(1-n), in SE 12.457 vs 12.5	0.5936 SE	3 SE	PASS
count variance / Hawkes (1971) closed form, 0.25 y Fano 5.93	1.019	0.15	PASS
MLE plant-and-recover, worst z	1.004	3	PASS
time rescaling KS p, Hawkes fit (should pass)	0.5838	0.01	PASS
time rescaling KS p, Poisson fit to the same events clustering is detected, not assumed	0	1.00e-06	PASS
second spike: superposition identity, 4 hosts inc2 - inc1 = r(d+W) - r(d)	8.88e-16	1.00e-12	PASS
Heston: property switches on at alpha = kappa (1.03 kappa) at 0.97 kappa: -2.1e-06	1.38e-06	0	PASS
rough (driver jumps): pairs with the property at n = 0.6 14 of 21 at n = 0.95 — needs near-critical clustering	0	0	PASS
kurtosis vs scale-mixture closed form, in SE 5.73 vs 5.59; ratio to Poisson = daily Fano	0.7597 SE	4 SE	PASS

Kalman

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check	measured	threshold	verdict
fast scalar path vs generic matrix filter	0	1.00e-12	PASS
RMSE improvement minus steady-state theory 45.6% vs 45.2%	0.3392 pts	2 pts	PASS
lecture AR(1) kappa on noisy quotes / true kappa attenuation bias; MLE through the filter is unbiased	12.6 x	2 x	PASS
dual vs joint EKF agreement	0.129 %	1 %	PASS
recursive MLE (Ljung) distance from the MLE Wan-Nelson dual: 0.07 SE	0.2287 SE	1 SE	PASS
strict: steps to re-converge after the regime change bad-print excursion 2.0x normal error	18	2	PASS
adaptive: bad-print excursion (x normal error) re-converges in 0 steps	14.92 x	6.044 x	PASS
robust: steps to re-converge (and ignores the print) excursion 1.7x vs strict 2.0x	1.5	2	PASS
lifted rough filter: RMSE improvement over quotes 24 lifted factors as the state, rank-one Q	47.41 %	30 %	PASS

Engineering

check	measured	threshold	verdict
scipy modules on the live import path scipy.stats alone costs 5.8 s	0	0	PASS
startup to actionable error, no TWS was 9.6 s before lazy imports	0.5385 s	3 s	PASS
calibration objective evaluation 39 quotes; 470 ms before vectorising	8.288 ms	200 ms	PASS
Gauss-Legendre rules constructed 2 reuses; leggauss is $O(n^2)$	1	1	PASS
quadrature tolerance 1e-12 vs 1e-14 calibration runs at 1e-12, ~2x faster	0	1.00e-12	PASS
replay round-trip fidelity recorded vs replayed surface points	0	1.00e-12	PASS

90 of 90 checks pass (29s).

Key results

Session D — the lift is a generalisation of Heston

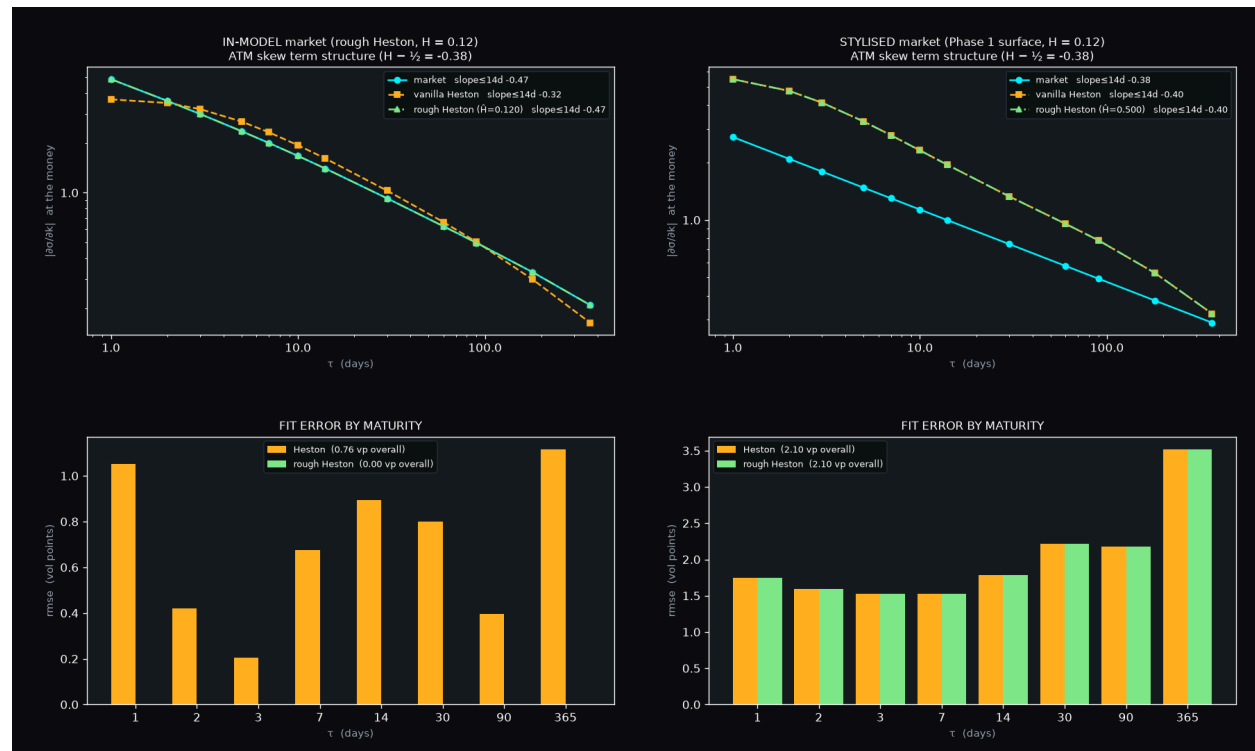
With one node at $x = 0$ the lift reproduces closed-form Heston to 6e-11 at half a year and 7e-10 at two years. The full 24-node lift converges to Heston **linearly** in $\frac{1}{2} - H$: 1.6e-3, 1.6e-4 and 1.6e-5 at $H = 0.49, 0.499$ and 0.4999 . The lifted recursion equals the convolution with the same sum-of-exponentials kernel to 5e-14. Against the true fractional kernel, from the same Brownian increments, the gap is 0.80% in L^2 . The lifted Monte Carlo agrees with the lifted characteristic function at $|z| < 0.4$.

Session E — rough Heston against vanilla Heston

Both models were fitted, by the same Levenberg–Marquardt driver with equal weights, to a market generated by rough Heston at $H = 0.12$ (8 maturities $\times$ 7 strikes).

	market	rough Heston	vanilla Heston
fit rmse	—	0.000 vp , all six parameters exact	0.76 vp at $\kappa = 35.6, \xi = 3.93$
one-day ATM skew	-4.84	-4.84 (100%)	-3.66 (76%)
log-log skew slope, 1–3 days	-0.441	-0.441	-0.123
log-log slope, 7–14 days	-0.512	-0.512	-0.525

Vanilla Heston copies the market over a fortnight: it pushes $1/\kappa$ to about ten days, so its transition from flat to decaying sits inside the window. **It cannot follow below a week**, where a diffusion's skew must go flat. Over the whole 1–14 day window the slope hides this (-0.32 against -0.47); the bend shows it. The market's own slope is -0.47, not $H - \frac{1}{2} = -0.38$: the power law is asymptotic, and at $\xi = 0.5$ the higher-order vol-of-vol term steepens it (-0.42 at $\xi = 0.05$).



Identifiability of H . Measured as a standard error, not assumed:

surface	weighting	SE(H)
4 expiries $\times$ 5 strikes	vega ² from a price half-spread, 0.5 vp noise	0.202 (2.3 effective quotes)
same	equal in vol	0.047
8 $\times$ 7 with a real short end	vega ² , noise consistent with a realistic spread	0.020

Under vega² weights the short end, which is where H lives, carries essentially no weight. Three noisy seeds gave $H = 0.50, 0.25$ and 0.12 , and in every case the optimiser had found a *lower* cost than the truth. $\text{corr}(H, \xi) = +0.98$: the rough analogue of the κ/θ trade-off in Session C.

Session F — Hawkes arrivals on three hosts

Your second-spike claim, made exact. By linearity, $E[\text{increment after shock 2}] - E[\text{increment after shock 1}] = r(d + w) - r(d)$, where r is the single-shock mean response (verified to $1e-15$). The second spike is larger exactly when r is still rising across the window.

- **Poisson, any host:** never. The response only decays. The *level* after the second shock is still higher by superposition, so a level comparison tests nothing.
- **OU and Heston with Hawkes:** at short gaps iff $\alpha > \kappa$, since $r'(0) = \eta(\alpha - \kappa)$. Confirmed: on at 1.03κ , off at 0.97κ .
- **Rough Heston, jumps through the Volterra kernel:** holds at **0 of 21** (gap, window) pairs at branching ratio 0.6, and at 14 of 21 at 0.95. A jump's effect decays like $t^{(H-1/2)}$, faster than excitation builds, so it needs near-critical clustering.
- **Rough Heston, jumps added directly to V:** holds at short gaps.

Which rough variant is right is a modelling choice; both are implemented.

Kurtosis. With constant variance and zero-mean Gaussian jumps, excess kurtosis is exactly $3s^4\text{Var}[N]/(vw + E[N]s^2)^2$. The Hawkes/Poisson ratio is therefore the daily Fano factor, 1.92. Simulation matches both closed forms, and on every stochastic-variance host Hawkes exceeds Poisson by 12–14 SE.



Session F — Kalman filtering

- **Against closed forms.** Matches your four-line recursion to $1.7\text{e-}16$. The gain converges to the analytic steady state, and the RMSE gain over raw quotes (45.6%) is what steady-state theory predicts (45.2%).
- **The lecture's AR(1) calibration is biased by quote noise.** κ comes out at 61–65 for a true 5, because noise attenuates the regression slope. Maximum likelihood through the filter recovers it within 1 SE.

Strict against adaptive, your two panels made numerical (40 seeds):

host	strict	adaptive	robust (persistence-gated)
OU	18 steps; bad print 2.3×	0 steps; 17×	1 step; 1.7×
Heston	2 steps; 3.2×	0 steps; 10×	1 step; 1.9×
rough	0 steps; 2.9×	0 steps; 6.2×	0 steps; 3.0×

On OU and Heston the persistence gate removes the trade-off. On rough Heston there is little regime to catch: a single shock relaxes like a power law.

Online parameter learning. Wan–Nelson's dual EKF and the joint EKF agree with each other to 0.14%, and both carry **Ljung's bias**: they ignore how the gain depends on the parameters, and land up to 2 SE from the MLE. The recursive MLE with full sensitivity equations stays within 1 SE on every host-seed (total deviation 2.5 SE against 4.6 SE).

Two rough-host findings.

- **Positivity floor.** QML κ is biased by 2–4 SE when simulated variance spends ~13% of days below zero, and unbiased ($z = +0.4$, $+0.4$) at parameters where it never does. The clamped drift is a non-linearity the linear filter cannot represent.
- **Forecasting.** A CIR filter fitted by MLE to rough variance forecasts it at 5 days within -0.6% to $+2.8\%$ of the *true* lifted model, by pushing κ to 43–54: Session E's phenomenon again. The lift's value is the correct state space, not a large short-horizon forecasting edge.

Corrections

1. **Session D claimed the implicit exponential-trapezoidal scheme is unconditionally stable.** It is not. At $u = 1200$, $H = 0.12$ and 90 days, 120 steps return $|\varphi| = 1.2e36$. The claim came from tests that only probed each maturity's pricing range. Stability constants are now measured for both schemes.
2. **Phase 1's "Heston delivers 58% of the one-day skew" depends on the weighting.** With vega² weights the one-day quotes barely enter the fit; with equal weights, vanilla Heston fitted to the same stylised surface delivers 201%. What survives is the misfit, not its sign. The stylised surface also cannot discriminate the models: rough Heston's best fit to it is $H = 0.5$ from every start. Annotated in `docs/phase1_baseline.md`.
3. **Plan targets that were not attainable at their stated sample sizes.**
 - H to ± 0.03 on a 20-quote vega²-weighted surface: SE 0.20.
 - κ within 10% after 2,000 filter steps: SE 20% of κ .
 - The Laplace-domain kernel error below 1% on $z \in [0.1, 100]$: 3.5%, because $z = 0.1$ probes ten years of memory.

Each is now tested against its measured standard error or documented.
4. **The plan's "adaptive-R" filter.** Inflating R after surprises *lowers* the gain; your lecture's adaptive filter is a dynamic *gain*. Corrected, and tested on both sides of the trade-off.
5. Smaller, caught by their own checks before they reached a result:
 - aliased maturities in the Session E capture;
 - spike windows overlapping the second shock, caught by the superposition identity;
 - an OU-host jump size that produced an excess kurtosis of 10,753;
 - the health-check RMSE trend rule, written 100× too loose.

Open flags

	severity	item
1	major	The live IBKR path has never run against a real TWS. Seven callbacks are exercised only by stubs.
2	major	eSSVI is not built; calendar arbitrage across slices is measured but not prevented.
3	major	Rough variance simulated by the lifted Euler scheme goes below zero (~13% of days at $\xi = 0.3$, $\theta = 0.04$). It biases QML parameter estimates and needs a positivity-preserving scheme.
4	moderate	H cannot yet be learned by the filter: it defines the lifted factors themselves.
5	moderate	A rough fit costs ~0.6 s per ten-expiry objective evaluation, so a multi-start calibration takes minutes. Fine for research, slow for live recalibration.
6	moderate	The stylised synthetic surface is not a stochastic-volatility smile. Future comparisons should use in-model markets or recorded real chains.
7	moderate	<code>volatility_surface_3.py</code> is 963 lines holding six classes and should be split.
8	minor	The lift's kernel is 12% off below five minutes; irrelevant at daily horizons, relevant intraday.
9	minor	The test runner is hand-rolled: no pytest, no CI, no coverage.

Resolved since the last report: the $\xi \rightarrow 0$ precision flag, the `fd_second` order-loss flag, and the Session D schedule risk.

Next

For your decision:

- **Real data.** Record IBKR chains with the replay source, and run the calibration and the filters on them.
- **Model fidelity.** Build a positivity-preserving rough variance scheme, which removes flag 3 and the QML bias with it.
- **Make roughness itself filterable.** Learn H in the filter.
- **Engineering.** Move the ten suites under pytest and CI.